Arithematic Progressions

Introduction:

A sequence is an arrangement of numbers according to definite value or pattern.

for eg:- 1, 3, 5, 7, 9 ...
1, 5, 10, 15, 20 ...
2, 0, -2, -4, -6 ...

The numbers occuring in a sequence are called in terms.

Arithematic Progressions

An arithematic progression is a list of no. in which each term is obtained by adding a fixed number to the proceeding term except the first term.

The first term of AP is denoted by small letter a.
The common difference is denoted by small letter d.

$a_n = a + (n-1)d$

The common difference is a fixed no. which can be positive, negative or zero.

d = 0

Ex- 1, 1, 1, 1

Finite AP

If the no. of term are countable in an AP then, AP is finite.

Non-finite AP / Infinite AP

If the no. of term are uncountable in an AP, then AP in non-finite / Infinite.

Ex- 4, 8, 12, 16, 20 ...

common difference (d) = $a_2 - a_1$

Exercise - 5.1

4-ii-Sol) The given terms are $2, \frac{5}{2}, 3, \frac{7}{2}, \ldots$

Here, $a_1 = 2$, $a_2 = \frac{5}{2}$,

$a_3 = 3$, $a_4 = \frac{7}{2}$.

Now, $a_2 - a_1 = \frac{5}{2} - 2 = \frac{1}{2}$;

$a_3 - a_2 = 3 - \frac{5}{2} = \frac{1}{2}$;

$a_4 - a_3 = \frac{7}{2} - 3 = \frac{1}{2}$; and so on

Hence, common difference is $\frac{1}{2}$

And, $a_5 = a + 4d = 2 + 4\left(\frac{1}{2}\right) = 4$

$a_6 = a + 5d = 2 + 5\left(\frac{1}{2}\right) = \frac{9}{2}$

$a_7 = a + 6d = 2 + 6\left(\frac{1}{2}\right) = 5$

Hence, Three more terms are $4, \frac{9}{2}, 5$.

v-Sol) The given terms are:

$3, 3+\sqrt{2}, 3+2\sqrt{2}, 3+3\sqrt{2}, \ldots$

Here, $a_1 = 3$, $a_2 = 3+\sqrt{2}$,

$a_3 = 3+2\sqrt{2}$, $a_4 = 3+3\sqrt{2}$,

Now, $a_2 - a_1 = 3+\sqrt{2} - 3 = \sqrt{2}$

$a_3 - a_2 = 3+2\sqrt{2} - 3 - \sqrt{2} = \sqrt{2}$

$a_4 - a_3 = 3+3\sqrt{2} - 3 - 2\sqrt{2} = \sqrt{2}$

Hence, common difference is $\sqrt{2}$

And, $a_5 = a + 4d = 3 + 4(\sqrt{2}) = 3 + 4\sqrt{2}$

$a_6 = a + 5d = 3 + 5(\sqrt{2}) = 3 + 5\sqrt{2}$

$a_7 = a + 6d = 3 + 6(\sqrt{2}) = 3 + 6\sqrt{2}$

Hence, Three more terms are $3+4\sqrt{2}$, $3+5\sqrt{2}$ and $3+6\sqrt{2}$.

xii-Sol) The given terms are $\sqrt{2}, \sqrt{8}, \sqrt{18}, \sqrt{32}, \ldots$

Here, $a_1 = \sqrt{2}$, $a_2 = \sqrt{8}$,

$a_3 = \sqrt{18}$, $a_4 = \sqrt{32}$, and so on

Now, $a_2 - a_1 = \sqrt{8} - \sqrt{2} = 2\sqrt{2} - \sqrt{2} = \sqrt{2}$,

$a_3 - a_2 = \sqrt{18} - \sqrt{8} = 3\sqrt{2} - 2\sqrt{2} = \sqrt{2}$,

$a_4 - a_3 = \sqrt{32} - \sqrt{18} = 4\sqrt{2} - 3\sqrt{2} = \sqrt{2}$,

Hence, common difference is $\sqrt{2}$

and, $a_5 = a + 4d = \sqrt{2} + 4(\sqrt{2}) = 5\sqrt{2}$

$a_6 = a + 5d = \sqrt{2} + 5(\sqrt{2}) = 6\sqrt{2}$

$a_7 = a + 6d = \sqrt{2} + 6(\sqrt{2}) = 7\sqrt{2}$

Hence, three more terms are $5\sqrt{2}$, $6\sqrt{2}$ and $7\sqrt{2}$.

xv-sol) The given terms are $1^2, 5^2, 7^2, 73, \ldots$

Here, $a_1 = 1^2 = 1$, $a_2 = 5^2 = 25$,

$a_3 = 7^2 = 49$, $a_4 = 73$, And so on

Now, $a_2 - a_1 = 25 - 1 = 24$

$a_3 - a_2 = 49 - 25 = 24$

$a_4 - a_3 = 73 - 49 = 24$

Hence, the common difference is 24

And, $a_5 = a + 4d = 1 + 4(24) = 97$

$a_6 = a + 5d = 1 + 5(24) = 121$

$a_7 = a + 6d = 1 + 6(24) = 145$

Hence, three more terms are 97, 121 and 145.

Exercise-5.2

3-iii-sol) Here, $a_1 = a = 5$

and $a_4 = a_{\frac{1}{2}}$

$\Rightarrow a + 3d = a_{\frac{1}{2}}$

$\Rightarrow 5 + 3d = \frac{19}{2}$

$\Rightarrow 10 + 6d = 19$

$\Rightarrow 6d = 9$

$\Rightarrow d = \frac{3}{2}$

Hence, $a_2 = a + d = 5 + \frac{3}{2} = \frac{13}{2}$

$a_3 = a + 2d = 5 + 2\left(\frac{3}{2}\right) = 8$

iv-sol) Here, $a_1 = -4$

And $a_6 = 6$

$a + 5d = 6$

$-4 + 5d = 6$

$5d = 10$

$d = 2$

Hence, $a_2 = a + d = -4 + 2 = -2$

$a_3 = a + 2d = -4 + 4 = 0$

$a_4 = a + 3d = -4 + 6 = 2$

$a_5 = a + 4d = -4 + 8 = 4$

4-Sol) Here, $a_1 = 38$

and $a_6 = -22$

$\Rightarrow a+d = 38$ — ①

and $a+5d = -22$ — ②

Subtracting ① from ②

$4d = -22-38$

$4d = -60$

$d = -15$

Hence, $a+(-15) = 38$

$a - 15 = 38$

$a = 53$

$a_3 = a+2d = 53+2(-15) = 23$

$a_4 = a+3d = 53+3(-15) = 8$

$a_5 = a+4d = 53+4(-15) = -7$

5-Sol-ii) The give A.P. is $18, 15\frac{1}{2}, 13, \ldots, -47$.

Here, $a_1 = 18$, $a_2 = 15\frac{1}{2} = \frac{31}{2}$, $a_3 = 13$

Now, $a_1 - a_1 = \frac{31}{2} - 18 = -\frac{5}{2}$

$\therefore a_3 - a_2 = 13 - \frac{31}{2} = -\frac{5}{2} = d$

$\therefore a_n = a+(n-1)d$

$\Rightarrow -47 = 18 + (n-1)\left(-\frac{5}{2}\right)$

$\Rightarrow -129 = -\frac{5n}{2} + \frac{5}{2}$

$\Rightarrow n = (-68)\frac{-2}{5}$

$\Rightarrow n = 26+1 = 27$

Hence, number of terms is 27.

6-Sol) The given A.P. is: $11, 8, 5, 2, \ldots$

Here, $a_2 - a_1 = 8-11 = -3$

$a_3 - a_2 = 5-8 = -3$

$a_4 - a_3 = 2-5 = -3 = d$

$\therefore a_n = a+(n-1)d$

$\Rightarrow -150 = 11+(n-1)(-3)$

$\Rightarrow -161 = (n-1)(-3)$

$\Rightarrow n-1 = \frac{-161}{-3}$

$\Rightarrow n = \frac{164}{3}$

Hence, -150 is not a term of the given A.P.

7-Sol) Let 'a' and 'd' be the first term and common difference respectively of the given A.P.

Given: $a_{11} = 38$

$a_{16} = 73$

$\Rightarrow a+10d = 38$ — ①

$\Rightarrow a+15d = 73$ — ②

Subtracting ① from ②

$\Rightarrow 5d = 35$

$\Rightarrow d = 7$

$\therefore a+10(7) = 38$

$a+70 = 38$

$a = -32$

$\therefore a+30d = a_{31}$

$-32+30(7) = a_{31}$

$-32+210 = a_{31}$

Hence, $a_{31} = 178$

9-Solu Let 'a' and 'd' be the first term and common difference respectively of the given A.P.

Given: $a_3 = 4$

and $a_9 = -8$ — (1)

$\Rightarrow a + 2d = 4$ — (2)

$\Rightarrow a + 8d = -8$

Subtracting (1) from (2)

$\Rightarrow 6d = -12$

$d = -2$

$\Rightarrow a + 2(-2) = 4$

$\Rightarrow a - 4 = 4$

$\Rightarrow a = 8$

$\Rightarrow 0 = 54 + (n-1)(-2)$

$\Rightarrow 8 - 2n + 2 = 0$

$\Rightarrow 10 = 2n$

Hence, $\Rightarrow n = 5$

11-Solu Let 'a' and 'd' be the first term and common difference respectively,

Here, $a_1 = 3$, $a_2 = 15$, $a_3 = 27$, $a_4 = 39$

Now, $a_2 - a_1 = 15 - 3 = 12$

$a_3 - a_2 = 27 - 15 = 12$

$a_4 - a_3 = 39 - 27 = 12$

$d = 12$

$a_{54} = 3 + (54-1)12$ & $a_n = 3 + (53)12 + 132$

$a_{54} = 3 + (53)12$ $\Rightarrow a_n = 636 + 132 + 3$

$a_{54} = 3 + 636$ $\Rightarrow 3 + (n-1)12 = 768 + 3$

$a_{54} = 639$ $\Rightarrow n - 1 = \frac{768}{12}$

$a_n = 3 + (n$ $\Rightarrow n = 65$

Hence, 65th term is 132 more than 54th term.

14-Solu $\Rightarrow a_1 = 12$ $\Rightarrow a_2 = 16$

$\Rightarrow a_{11} = 16$ $\Rightarrow a_4 = 24$

$\Rightarrow a_2 - a_1 = 16 - 12$

$= 4$

$\Rightarrow a_3 - a_2 = 20 - 16$

$= 4$

$\Rightarrow a_4 - a_3 = 24 - 20$

$= 4$

$d = 4$

$\Rightarrow a_n = 248$

$\Rightarrow 248 = 12 + (n-1)4$

$\Rightarrow 236 = 4n - 4$

$\Rightarrow 240 = 4n$

$\Rightarrow n = 60$

Hence, 60 terms lie between 10 and 250.

17-Solu $\Rightarrow a_1 = 3$ $\Rightarrow a_2 = 13$

$\Rightarrow a_2 = 8$ $\Rightarrow a_n = 253$

$\Rightarrow a_2 - a_1 = 8 - 3 = 5$

$\Rightarrow a_3 - a_2 = 13 - 8 = 5$

$\Rightarrow d = 5$

Now, $a_n = 253$

$a + (n-1)d = 253$

$3 + (n-1)(5) = 253$

$n - 1 = 50$

$n = 51$

20th term from the end
= Total number of terms - 20 + 1
= 51 - 20 + 1 = 32

$\therefore a_{32} = 3 + (31)(5)$
$= 3 + 155 = 158$

Hence, the 20th term from the end is 158.

18-Solu → $a_4 + a_8 = 24$
$\Rightarrow a + (4-1)d + a + (8-1)d = 24$
$\Rightarrow a + 3d + a + 7d = 24$
$\Rightarrow 2a + 10d = 24$
$\Rightarrow a + 5d = 12$ — ①

and $a_8 + a_{10} = 44$
$\Rightarrow a + (8-1)d + a + (10-1)d = 44$
$\Rightarrow 2a + 14d = 44$
$\Rightarrow a + 7d = 22$ — ②

Subtracting ① from ②,
$2d = 10$
$d = 5$

Putting in ①,
$a + 5(5) = 12$
$a = 12 - 25$
$a = -13$

Hence, the first three terms of the AP are:
$-13, -13+5$ and $-13+2(5)$ i.e., $-13, -8$ and -3.

19-Solu Subba Rao's starting salary = ₹5000
Annual increment = ₹200
Thus, 'a' = ₹5000
and, 'd' = ₹200

Let the salary reach ₹7000 in n years
$\therefore a_n = ₹7000$
$a + (n-1)d = ₹7000$
$\Rightarrow 5000 + (n-1)(200) = 7000$
$\Rightarrow (n-1)(200) = 7000 - 5000$
$\Rightarrow (n-1)(200) = 2000$
$\Rightarrow n - 1 = 10$
$\Rightarrow n = 11$

For years, the sequence is 1995, 1996, 1997...
Here, $a = 1995$,
$d = 1$
$n = 11$
$a_n = 1995 + (11-1)(1)$
$= 1995 + 10 = 2005$

Hence, in 2005, Subba Rao's salary was ₹7000.

Ex-5.3

1-i-Sol: AP: 2, 7, 12, ...

$a = 2$, $d = 7-2 = 5$

$n = 10$

$S_{10} = \frac{10}{2}(2(2) + (10-1)(5))$

$= 5(4+45) = 245$

ii-Sol: AP: $-37, -33, -29 \ldots$

$a = -37$, $d = -33-(-37)$

$= -33+37$

$= 4$

$n = 12$

$S_{12} = \frac{12}{2}(2(-37) + (12-1)(4))$

$= 6(-74+44) = 6(-30)$

$= -180$

2-ii-Sol: AP: $34+32+30+ \ldots +10$

$a = 34$

$d = 32-34 = -2$

$l = a_n = 10$

$10 = a + (n-1)d$

$10 = 34 + (n-1)(-2)$

$(n-1)(-2) = -24$

$n-1 = 12$

$n = 13$

$S_{13} = \frac{13}{2}(34+10) = \frac{13}{2} \times 44 = 13 \times 22 = 286$

iii-Sol: $AP = (-5) + (-8) + (-11) + \ldots + (-230)$

$a = -5$, $d = -8-(-5) = -3$

$l = a_n = -230$

$a + (n-1)d = -230$

$-5 + (n-1)(-3) = -230$

$n-1 = 75$

$n = 76$

$S_{76} = \frac{76}{2}(-5-230)$

$= 38(-235) = -8930$

3-vi-Sol: $\Rightarrow a = 2$, $d = 8$, $S_n = 90$

$\therefore \frac{n}{2}(2 \times 2 + (n-1)(8)) = 90$

$n(4+8n-8) = 180$

$n(2+4n-4) = 90$

$n(4n-2) = 90$

$4n^2 - 2n = 90$

$2n^2 - n = 45$

$2n^2 - n - 45 = 0$

$2n^2 - 10n + 9n - 45 = 0$

$2n(n-5) + 9(n-5) = 0$

$(2n+9)(n-5) = 0$

$n-5 = 0$

or $2n+9 = 0$

$n = 5$ or $n = -\frac{9}{2}$

Thus, $n = 5$ (n cannot be -ve)

and $a_n = a_5 = a + (n-1)d$

$= 2 + (4)(8)$

$= 34$

3rd Solu ⇒ a = 3, n = 8, S = 192

Now, S = 192

$S_8 = 192$

$\Rightarrow \frac{8}{2}(2\times 3 + (8-1)(d)) = 192$

$\Rightarrow 4(6+7d) = 192$

$\Rightarrow 6+7d = 48$

$\Rightarrow 7d = 42$

$\Rightarrow d = 6$

8th Solu ⇒ l = 28, S = 144 and n = 9

Since, $S_9 = 144$,

$\Rightarrow \frac{9}{2}(a+28) = 144$

$\Rightarrow a+28 = 32$

$\Rightarrow a = 4$

4th Solu AP : 9, 17, 25, ...

$a = 9 \quad d = 17-9 = 8$

Now, $S_n = 636$

$\Rightarrow \frac{n}{2}(2(9) + (n-1)(8)) = 636$

$\Rightarrow \frac{n}{2}(18+8n-8) = 636$

$\Rightarrow n(a+4n-4) = 636$

$\Rightarrow n(4n+5) = 636$

$\Rightarrow 4n^2 + 5n - 636 = 0$

$\therefore n = \frac{-b \pm \sqrt{D}}{2a}$

$= \frac{-5 \pm \sqrt{(5)^2 - 4(4)(-636)}}{2(4)}$

$= \frac{-5 \pm \sqrt{25 + 10176}}{8}$

$= \frac{-5 \pm \sqrt{10201}}{8}$

$= \frac{-5 \pm 101}{8} = \frac{-5+101}{8} = \frac{96}{8} = 12$

or

$= \frac{-5-101}{8} = \frac{-106}{8} = -\frac{53}{4}$

$n = 12$ (n cannot be -ve)

Hence, the sum of 12 terms is 636.

5th Solu $a = 5, \quad l = 45$

$S_n = 400$

Now, $l = a_n = 45$

$\Rightarrow a + (n-1)d = 45$

$\Rightarrow (n-1)d = 40 \quad -①$

and $S_n = 400$

$\frac{n}{2}(a+l) = 400$

$\frac{n}{2}(5+45) = 400$

$25n = 400$

$n = 16$

Putting in ①,

$\Rightarrow (16-1)d = 40$

$\Rightarrow 15d = 40$

$\Rightarrow d = \frac{40}{15} = \frac{8}{3}$

Hence, $n = 16$ and $d = \frac{8}{3}$

6-Sol) $a = 17$, $L = 350$ and $d = 9$

$L = a_n = 350$

$a + (n-1)(d) = 350$

$17 + (n-1)(9) = 350$

$n - 1 = 37$

$n = 38$

and $S_{38} = \frac{38}{2}(17+350)$

$= 19(367) = 6973$

Hence, The sum of 38 terms of given AP is 6973

7-Sol) $d = 7$, $a_{22} = 149$ and find $n = 22$

$a_{22} = 149$,

$a + (n-1)d = 149$

$a + (22-1)(7) = 149$

$a + (21)(7) = 149$

$a = 149 - 147 = 2$

and $S_{22} = \frac{22}{2}(a + a_{22}) = 11(2+149) = 1661$

Hence, sum of first 22 terms is 1661

10-i-Sol) Given: $a_n = 3 + 4n$ — ①

Putting $n = 1, 2, 3, \ldots$ in ①, we get:

$a_1 = 3 + 4(1) = 7$,

$a_2 = 3 + 4(2) = 11$,

$a_3 = 3 + 4(3) = 15, \ldots$; and so on.

Now, $a_2 - a_1 = 11 - 7 = 4$

and $a_3 - a_2 = 15 - 11 = 4$

Thus, a $d = 4$

Hence the given sequence forms an A.P.

Here, $a = 7$, $d = 4$ and $n = 15$

$S_{15} = \frac{15}{2}(2(7) + (15-1)(4))$

$= \frac{15}{2}(8+56) = 15 \times 35 = 525$

Hence, the sum of first 15 terms is 525

11-Sol) Given: $S_n = 4n - n^2$

Putting $n = 1$,

$S_1 = 4(1) - (1)^2 = 4 - 1 = 3$

$a = a_1 = S_1 = 3$

Putting $n = 2$,

$S_2 = 4(2) - (1)^2 = 8 - 4 = 4$

$a_1 + a_2 = 4$

$3 + a_2 = 4$

$a_2 = 1$

Putting $n=3$,

$S_3 = 4(3) - (3)^2 = 12 - 9 = 3$

$S_2 + a_3 = 3$

$4 + a_3 = 3$

$a_3 = -1$

Now, $d = a_3 - a_2 = -1 - 2 = -3$

$a_{10} = 3 + 9(-3) = 3 - 27 = -24$

$a_n = 3 + (n-1)(-3) = 3 - 3n + 3 = 6 - 3n$

Hence, $a_3 = -1$, $a_{10} = -24$

and $a_n = 6 - 3n$

12-Solu. First positive integers divisible by 6 are:

6, 12, 18, 24

Here, $a = a_1 = 6$, $a_2 = 12$,

$a_3 = 18$, $a_4 = 24$; and so on.

$\therefore$ $a_2 - a_1 = 12 - 6 = 6$

$a_3 - a_2 = 18 - 12 = 6$

$a_4 - a_3 = 24 - 18 = 6$

$\therefore$ $d = 6$

Now, $S_{40} = \frac{40}{2}(2(6) + (40-1)6)$

$= 20(12 + 234) = 4920$

Hence, the sum of first 40 positive integers divisible by 6 is 4920.

13-Solu. First multiples of 2 are 2, 16, 24, 32, ...

Here, $a = a_1 = 2$, $a_2 = 16$,

$a_3 = 24$, $a_4 = 32$; and so on.

$a_2 - a_1 = 16 - 2 = 8$

$a_3 - a_2 = 24 - 16 = 8$

$a_4 - a_3 = 32 - 24 = 8$

$\therefore$ $d = 8$

Now, $S_{15} = \frac{15}{2}(2 \times 2 + (14)8)$

$= \frac{15}{2}(4 + 112) = 960$

Hence, the sum of first 15 multiples of 2 is 960.

15-Solu. Let L_1 = Length of first semicircle

$= \pi r_1 = \pi(0.5) = \frac{\pi}{2}$

L_2 = Length of second semicircle

$= \pi r_2 = \pi(1) = \pi$

L_3 = Length of third semicircle

$= \pi r_3 = \pi(1.5) = \frac{3}{2}\pi$

and L_4 = length of fourth semi circle

$= \pi r_4 = \pi(2) = 2\pi$; and so on.

These lengths form an AP

Here, $a = a_1 = \frac{\pi}{2}$; $a_2 = \pi$;

$a_3 = \frac{3\pi}{2}$; $a_4 = 2\pi$ ...

and $n = 13$

$\therefore \quad d = a_2 - a_1 = \pi - \frac{\pi}{2} = \frac{\pi}{2}$

$\therefore$ Length of whole spiral $= S_{13}$

$$= \frac{13}{2}\left(2\left(\frac{\pi}{2}\right) + (13-1)\left(\frac{\pi}{2}\right)\right)$$

$$= \frac{13}{2}\left(\pi + (6)\left(\frac{\pi}{2}\right)\right)$$

$$= \frac{13}{2}(7\pi) = \frac{13}{2} \times \frac{7}{1} \times \frac{22}{7} = 143$$

Hence, total length of spiral made up of thirteen consecutive semicircles is 143 cm.

20-Sol. Distance covered to pick up the 1st potato
$= 2 \times 5m = 10m$

Since the distance between successive potatoes $= 3m$
$\therefore$ Distance covered to pick up the 2nd potato
$= 1(5+3)\,m = 16m$

Distance covered to pick up the 3rd potato
$= (5+3+3) = 22m$

These form an AP

Here, $a_1 = a = 10, \; a_2 = 16, \; a_3 = 22,$

$d = a_2 - a_1 = 16 - 10 = 6$

and $n = 10$

$\therefore$ Total distance covered $= S_{10}$

$$= \frac{10}{2}(2(10) + (10-1)(6))$$

$$= 5(20 + 54) = 370$$

Hence, total distance is 370 m which the competitor has to run.

17/8/21